Q.1 The differential equation $$\begin{array}{l} y(1 + 2xy)dx + x(1 - 2xy)dy = 0\\ \end{array}$$ is

A. None of these

B. Separable

C. Exact

D. Homogeneous

Q.2 $${\text{ydx - y(sinx)dy = 0, is an example of - - - differential equation}}{\text{.}}$$

A. Non-exact

B. Non-homogeneous

C. Exact

D. Non-linear

Q.3 Classify the following differential equation $${e^x}\frac{{dy}}{{dx}} + 2y = 3xy$$

A. Linear and not separable

B. Separable and not linear

C. Neither separable nor linear

D. Both separable and linear

Q.4 $$Which ~of ~the ~following ~substitution~ will~ transform~ the~ differential~ equation:\frac{{dy}}{{dx}} = \frac{y}{x} + \sec \left({\frac{y}{x}} \right),~in ~to ~separable~ form?$$

A. $$x = vy$$

B. $$y = vx$$

C. $$y = v + x$$

D. $$y = v - x$$

Q.5 $$For~ f(x,y) = \frac{{2x}}{{3y}} + 7,f(tx,ty) = - - -$$

A. $$f(x,y)$$

B. $${t^3}f(x,y)$$

C. $$tf(x,y)$$

D. $tf(x,y)$

Q.6 $$Integrating~factor~for~the~first~order~linear~differential~equation~:{\left({x - 1} \right)^3}\frac{{dy}}{{dx}} + 4{\left({x - 1} \right)^2}y = \cos~ x~is - - -$$

A. $${e^{x - 1}}$$

B. $${e^{{1}{{x - 1}}}}$$

C. $$ (x - 1)$$

D. $${(x - 1)^4}$$

Q.7 The differential equation $$\frac{{dy}}{{dx}} - y = {y^3}$$ may be called as linear equation of

A. Bessel

B. Cauchy

C. Homogeneous

D. Bernoulli's

Q.8 $$\begin{gathered} {\text{A differential equation M(x,y)dx + N(x,y)dy = 0 is exact if there exists a}} \hfill \\\ {\text{multi - variable function f(x,y) such that - - - }} \hfill \\\\ \end{gathered}$$

A. $$df({x,y}) = {{ f}}{{ x}}dx + {{ f}}{{ y}}dy$$

B. $$f(x,y) = {{{ f}}{{ x}}} dx + {{{ f}}{{ y}}} dy$$

C. $$f(x,y) = {{ f}}{{ x}}dx + {{ f}}{{ y}}dy$$

D. $$ f (x,y)dx = {{{ f}}{{ x}}} dx + {{{ f}}{{ y}}} dy$$

Q.9 The general linear ordinary differential equation of order n is:
$a_{n}(x)\frac{d^{n}y}{dx^{n}}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots +a_{2}(x)\frac{d^{2}y}{dx^{2}}+a_{1}(x)\frac{dy}{dx}+a_{0}(x)y=0$,$\,$where -------.

A. $a_{n}(x)=0$

B. $a_{n}(x) 0$

C. $a_{0}(x)=0$

D. $a_{0}(x) 0$

Q.10 For the nonexact differential equation $$y\left({xy + 2{x^2}{y^2}} \right)dx + x\left({xy - {x^2}{y^2}} \right)dy = 0,ifxM(x,y) - yN(x,y) = {x^3}{y^3} \ne 0,$$then the integrating factor is - - -

A. $${-1}{{{x^3}{y^3}}}$$

B. $${1}{{{x^3}{y^3}}}$$

C. $${e^{ - {x^3}{y^3}}}$$

D. $${e^{{x^3}{y^3}}}$$
